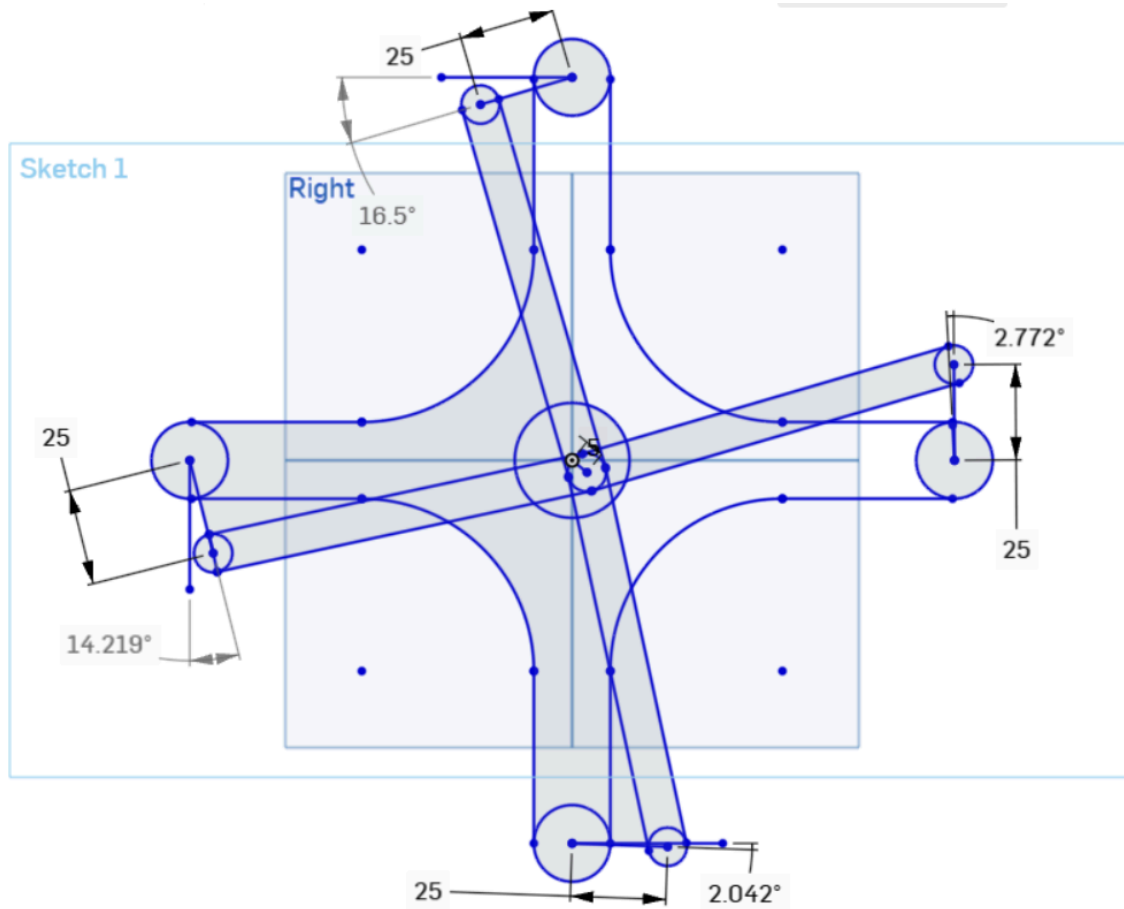
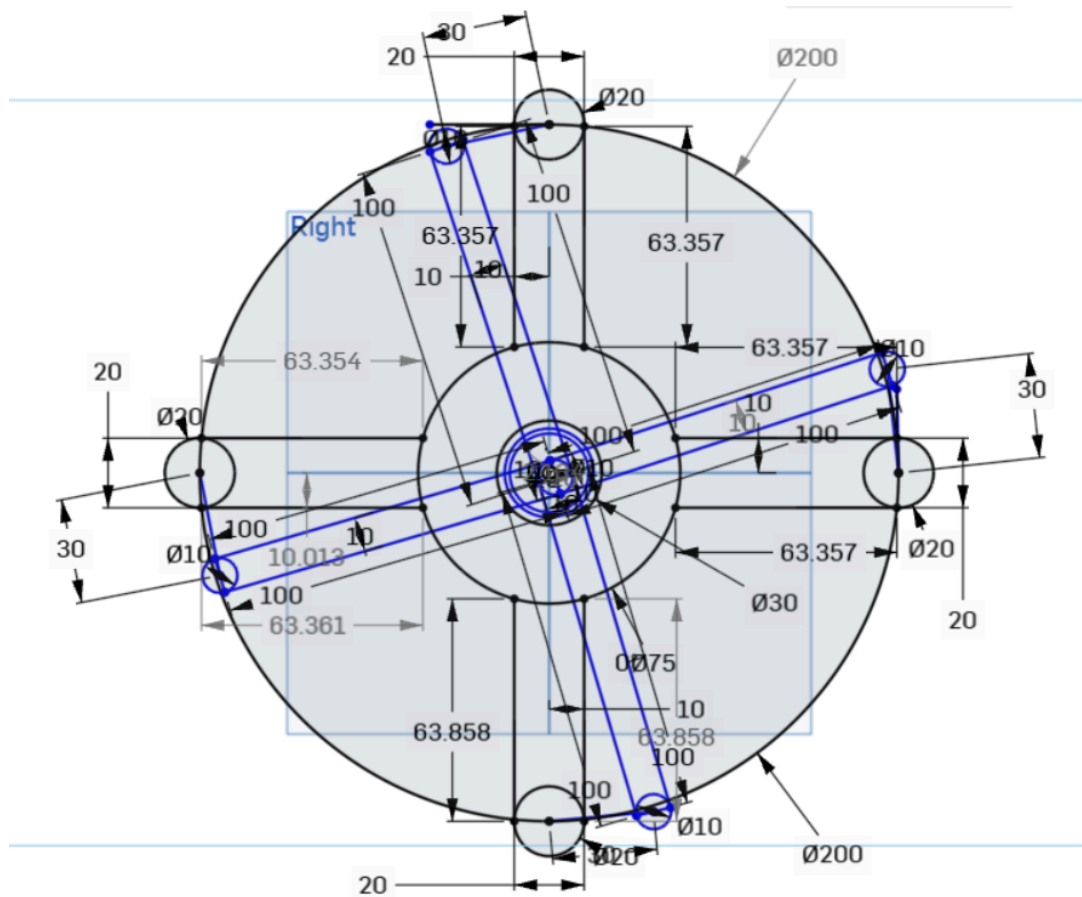


1. Executive Summary & Concept Overview

This proposal presents the preliminary design of a **4-blade barrel-type cycloidal rotor module** engineered to meet and exceed the PUSHPAK Grand Challenge 2026 targets. By integrating a belt-reduction drive, an eccentric cyclic pitch mechanism ($e = 5\text{ mm}$), and a strict mass optimization strategy, this design targets a sustained **12 N of thrust** while maintaining a complete-module mass under **350 g**, yielding an exceptional thrust-to-weight ratio (T/W) of **2.50**.





2. Preliminary Rotor Sizing & Aerodynamic Baseline

To guarantee compliance with competition minimums (10N thrust, 2.5 T/W) while embedding a **20% design contingency buffer**, the baseline geometry is defined below:

Parameter	Proposed value
Rotor diameter	200 mm
Rotor radius	100 mm
Blade span	200 mm
Number of blades	4
Blade chord	30 mm
Nominal RPM	4,500 RPM
Blade airfoil	Symmetric low-Re section; NACA 0012 as initial baseline

Pitch control	Cyclic pitch through eccentric/offset mechanism
Initial pitch amplitude sweep	$\pm 17^\circ$ (estimated)
Blade structure	Foam core + carbon spar + thin aerodynamic skin
Spar	Carbon-fiber tube/rod along span
Pitch links	Carbon-fiber rod + printed/metal end fittings
Frame	Lightweight 3D-printed structure with localized reinforcement
Drive	BLDC motor + belt reduction
Bearings	2 supported rotor-shaft bearings
Design thrust target	12 N
Minimum required thrust	10 N
Complete-module mass target	≤ 350 g
Absolute mass limit at 10 N	408 g
Target T/W at 12 N, 350 g	3.50
Target T/W at 10 N, 350 g	2.91

3. Blade Arrangement and Pitch-Control Concept

- **Mechanism Architecture:** Cyclic blade-pitch variation is executed through an **eccentric/offset control ring mechanism** ($e = 5$ mm).
- **Kinematic Function:** Each of the 4 blades pivots independently around its spanwise axis. Miniature carbon-fiber push-rods connect the blade pitch horns to the off-center control ring. As the rotor assembly spins at nominal 4,500 RPM, the offset geometry forces a precise sinusoidal pitch change, sweeping through an experimentally validated range of **-17 to +17 degrees**.
- **Thrust Vectoring Control:** Shifting the geometric center of the eccentric control ring (via dual high-speed micro-servos) alters the phase angle of maximum lift, instantly vectoring total thrust through 360 degrees without altering motor RPM.
- **Backlash Elimination:** Mechanical play is minimized by utilizing precision-toleranced carbon-fiber links with metal end fittings and a dual-bearing main shaft support.

4. Estimated Thrust and Power Requirement

- **Design Thrust Target: 12 N** (providing a 20% engineering safety cushion over the 10 N minimum requirement to absorb real-world wake and boundary-layer losses).
- **Power Estimation:** Powering the module via a high-torque brushless outrunner motor coupled with a **belt-reduction drive system** ensures efficient torque delivery at lower motor temperatures. Peak electrical power is estimated at **180–220 W** during high-vectoring maneuvers at 4,500 RPM.

5. Estimated Module Weight and Thrust-to-Weight Ratio

To satisfy the strict constraint of a T/W greater than 2.5, our absolute mass limit for a 10 N thrust output is 408g. Our design implements an aggressive weight reduction plan targeting a complete-module mass of **350g** (approx 14% safety buffer).

Component Breakdown	Estimated Mass
Rotor Blades (4x with internal carbon spars & skin)	60 g
Central Hub, End Spoke Plates & Main Shaft	95 g
Eccentric Control Ring, Pitch Links & Actuation Servos	50 g
BLDC Motor & Belt-Reduction Drive Assembly	105 g
Lightweight 3D-Printed Frame & Mounts	40 g
Total Complete-Module Mass Target	350 g

6. Initial Material and Manufacturing Approach

- **Blades:** High-modulus **structural foam core** wrapped in a thin, rigid aerodynamic skin, reinforced with a central pultruded **carbon-fiber tube/rod spar** along the 200 mm span to resist bending and torsional flutter under high centrifugal loads.
- **Pitch Links:** Pultruded **carbon-fiber rods** paired with high-strength precision metal/printed end fittings to eliminate elongation under load.
- **Frame & Structural Mounts:** **Lightweight 3D-printed engineering thermoplastics** (e.g., Nylon-Carbon Fiber composite) featuring localized topological ribs and structural webbing to withstand operational vibrations.
- **Drive & Bearings:** Driven via a robust BLDC motor and a timing-belt reduction setup, supported by a dual-bearing arrangement on the main rotor shaft to ensure coaxial stability.

7. Team Capability and Execution Plan

Core Competencies & Toolchain:

- **CAD & Mechanical Modeling:** Parametric modeling, constraint management, and assembly motion analysis leveraging cloud-native **Onshape** for rapid iteration and version control.
- **Simulation & Analysis:** Advanced multiphysics modeling using **COMSOL Multiphysics** to execute fluid-structure interaction (FSI), blade aerodynamic performance (CFD), and structural stress assessment (FEA).
- **Control & Embedded Systems:** Strong software foundation for developing the precise electronic speed control and micro-servo mixing algorithms required for real-time thrust vectoring.

Execution Roadmap (Stage 1 to Finale):

- **Phase 1 (Stage 1 - Sept 2026):** Preliminary design closure, baseline CAD layout finalization in Onshape, and theoretical thrust/mass optimization (Current Milestone).
- **Phase 2 (Stage 2 - Oct to Dec 2026):** Execution of detailed CAE analysis—running CFD and moving-mesh simulations in COMSOL on the NACA 0012 blade profile and performing FEA on the central hub and carbon spars. Finalization of the Bill of Materials (BOM) and electrical control architecture.
- **Phase 3 (Techfest Finale - Dec 2026):** Presentation of the complete manufacturing-ready design package, simulation telemetry, and control integration defense before the IIT Bombay evaluation committee.

8. References

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